BiomedCLIP

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Background

Background

Problem:
- Biomedical data are multimodal (text + image etc.)
- A general biomedical foundation model should be able to process this

Limitations (On dataset):
- Proprietary training data, private access
- Small dataset scale
- Lack diversity (Chest X-rays data only)

Background

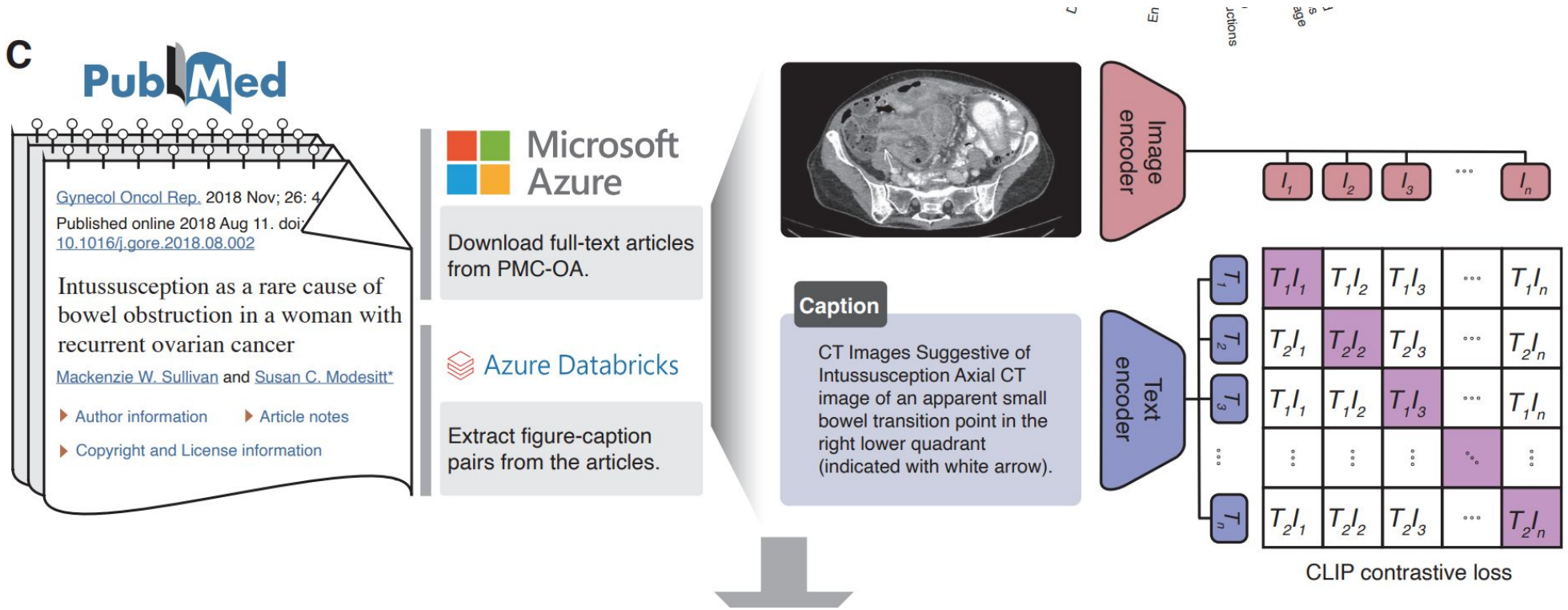
What They Do:

- Raise PMC-15M (15M image-text pairs from 4.4M scientific articles)
- 30+ major biomedical image types, clinical and research fields
- Open access

- Pretrained BiomedCLIP (Outperforms CLIP and PubMedCLIP, MedCLIP etc.)
- Excels in cross-modal retrieval, zero-shot image classification, and VQA
- Demonstrates robust performance when downstream data is scarce.

Methods

Methods – PMC-15M



PMC-15M Creation Procedure

Methods – Fine-Grained-46M

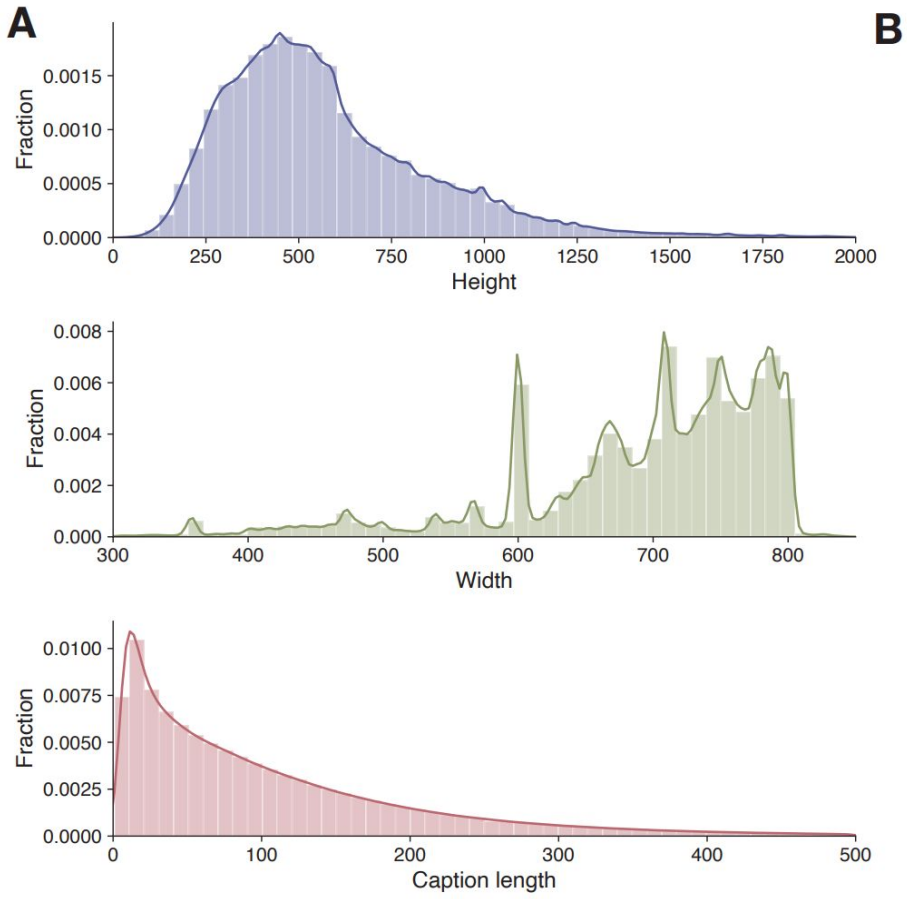
The Challenge: Compound Figures

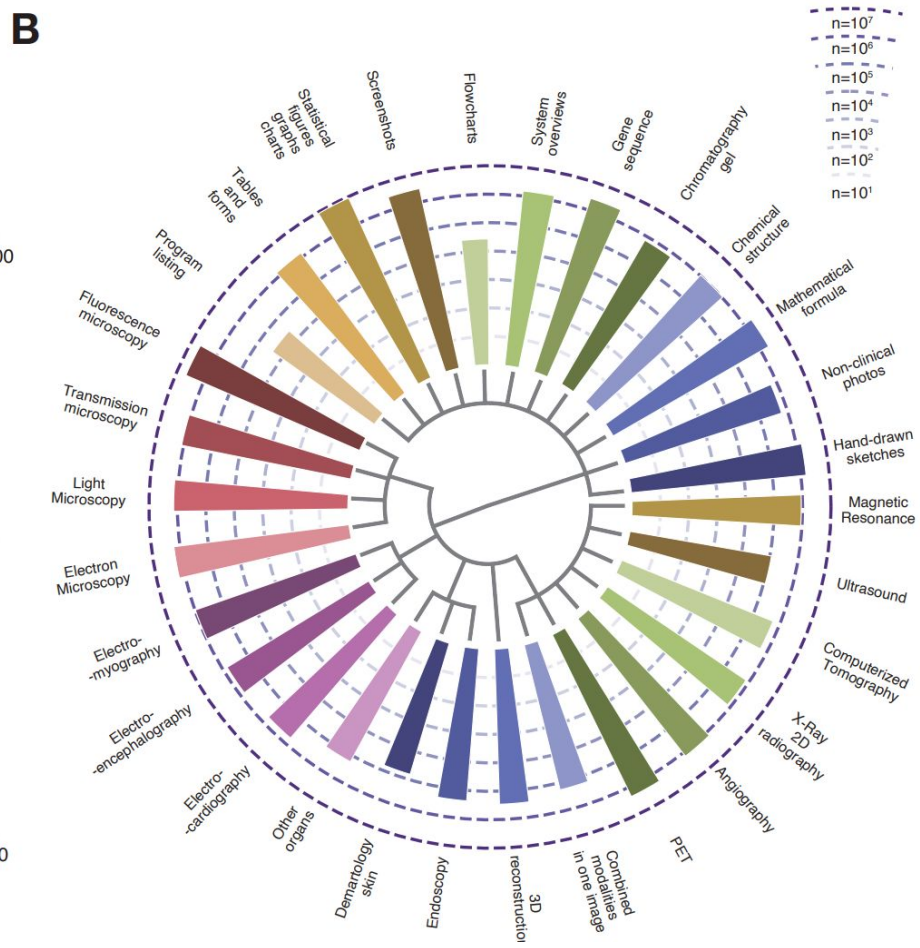
- **The "Eye-Blur" Effect:** Multiple panels (e.g., A, B, C, D) in one image
- **Semantic Noise:** Confuses model, poor feature alignment
- **The Goal:** Transition from "Overview" to "Pixel-Level Understanding"

The Automated Curation Pipeline

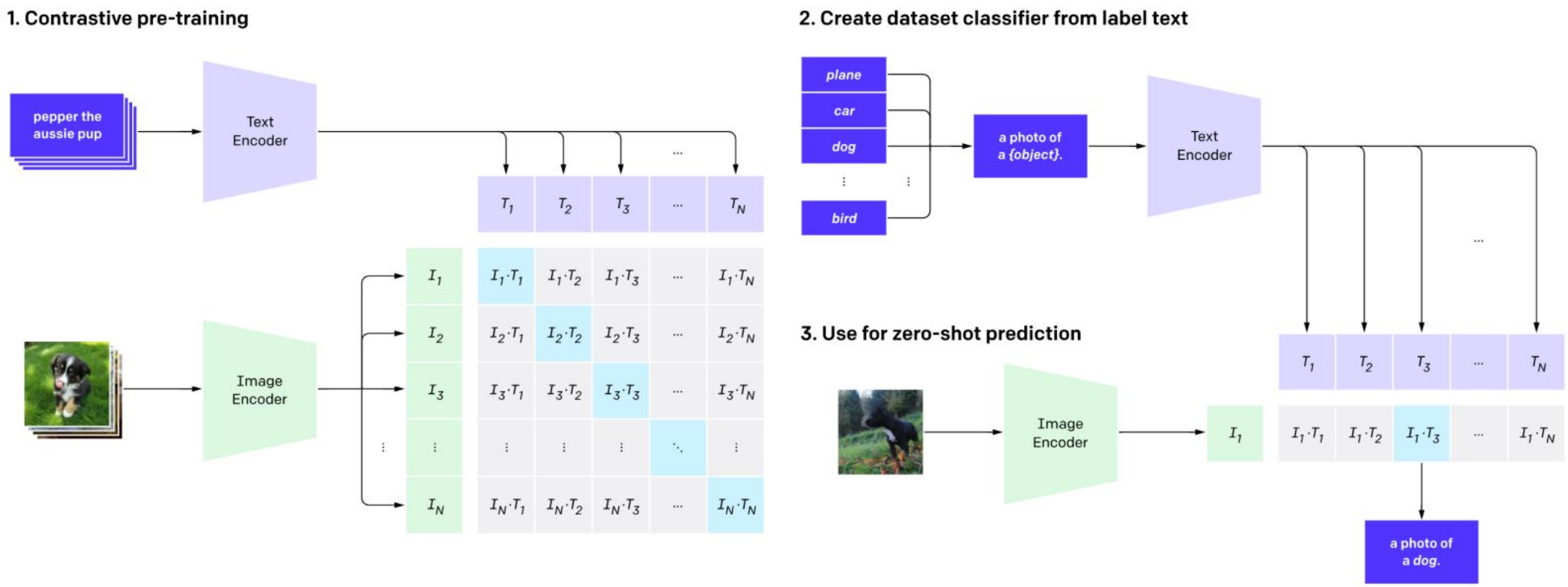
- **Citance Extractor:** In-line references (citances) for richer context
- **Caption Splitter:** Regex to sub-captions mapped to specific labels
- **OCR:** Detects sub-figure labels (e.g., "A", "b“) directly
- **Figure Splitter & Matcher:** Independent panels pairs with corresponding sub-captions

Methods - Dataset





Methods - BiomedCLIP



OpenAI

Methods – BiomedCLIP

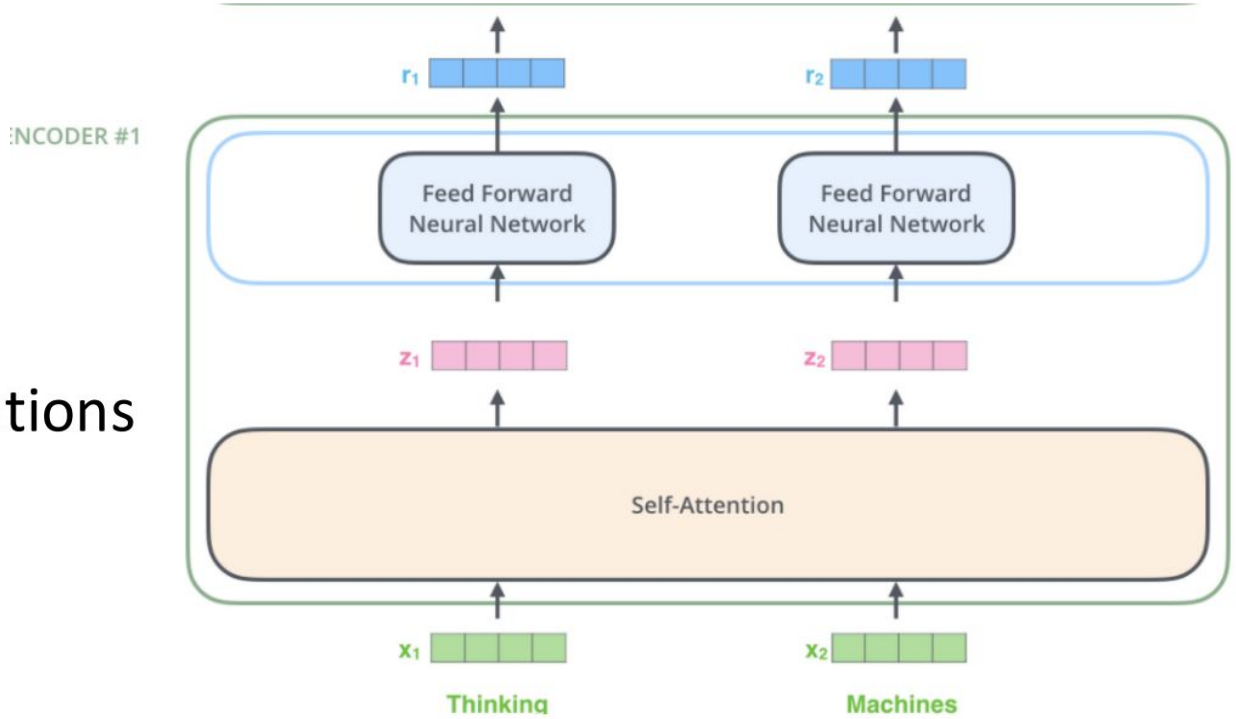
Changes from CLIP

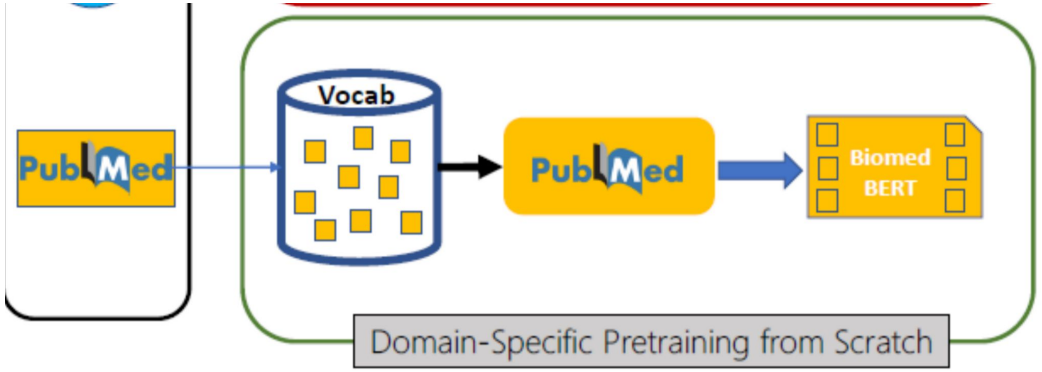
- **Text Encoder:** GPT-2 > PubMedBERT
- **Image Encoder:** ViT/ResNet > ViT + PatchDropout
- **Different Tokenizer:** BPE (Byte-Pair Encoding) > WordPiece (BioBERT Style)

  (Less vocabulary size but includes more medical contexts + longer max tokens)

Methods - BERT/PubMedBERT

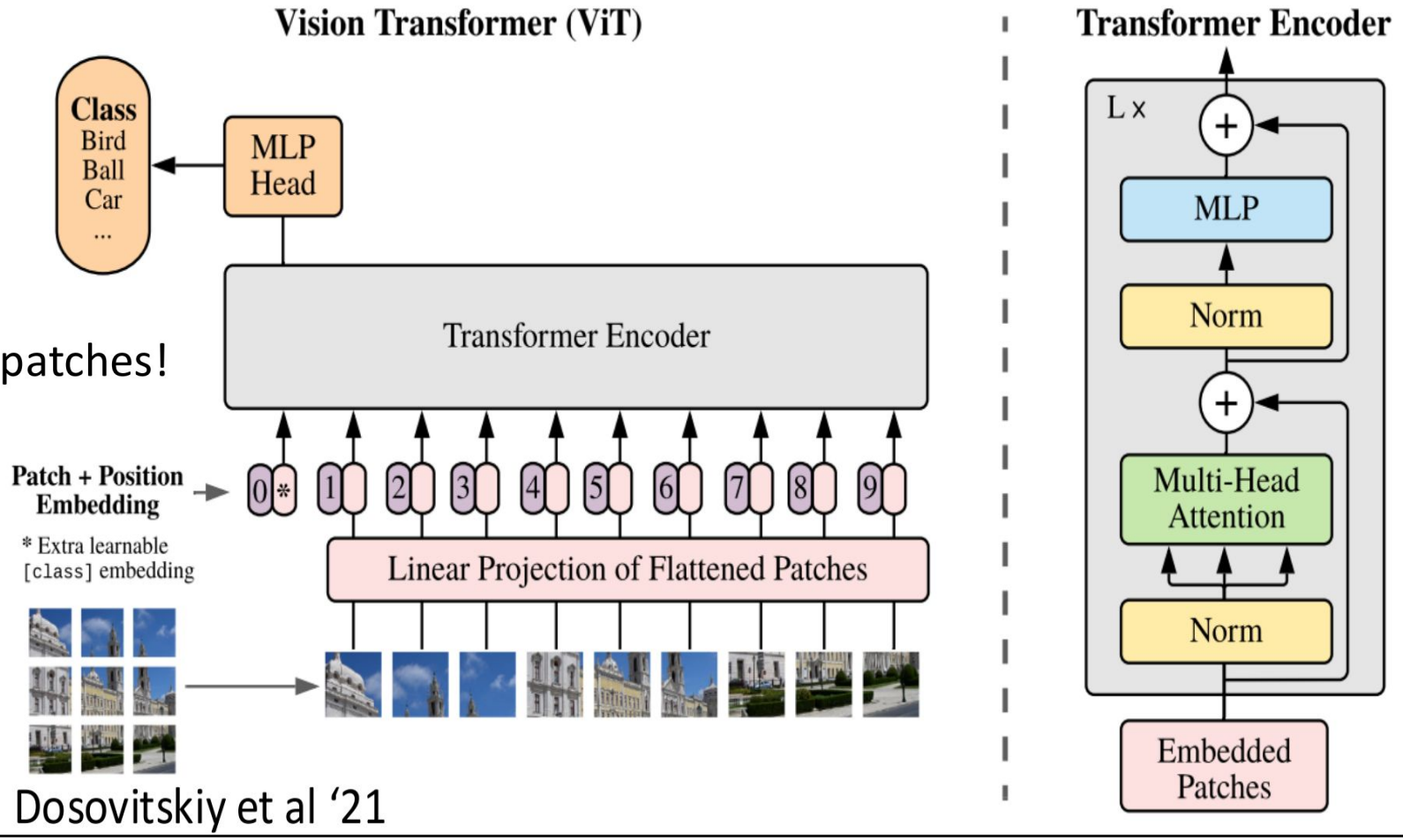
not a generative model → get representations





Methods - ViT

Can also use for vision: **ViT.** Just use patches!



Dosovitskiy et al '21

Methods - PatchDropout

High Spatial Redundancy

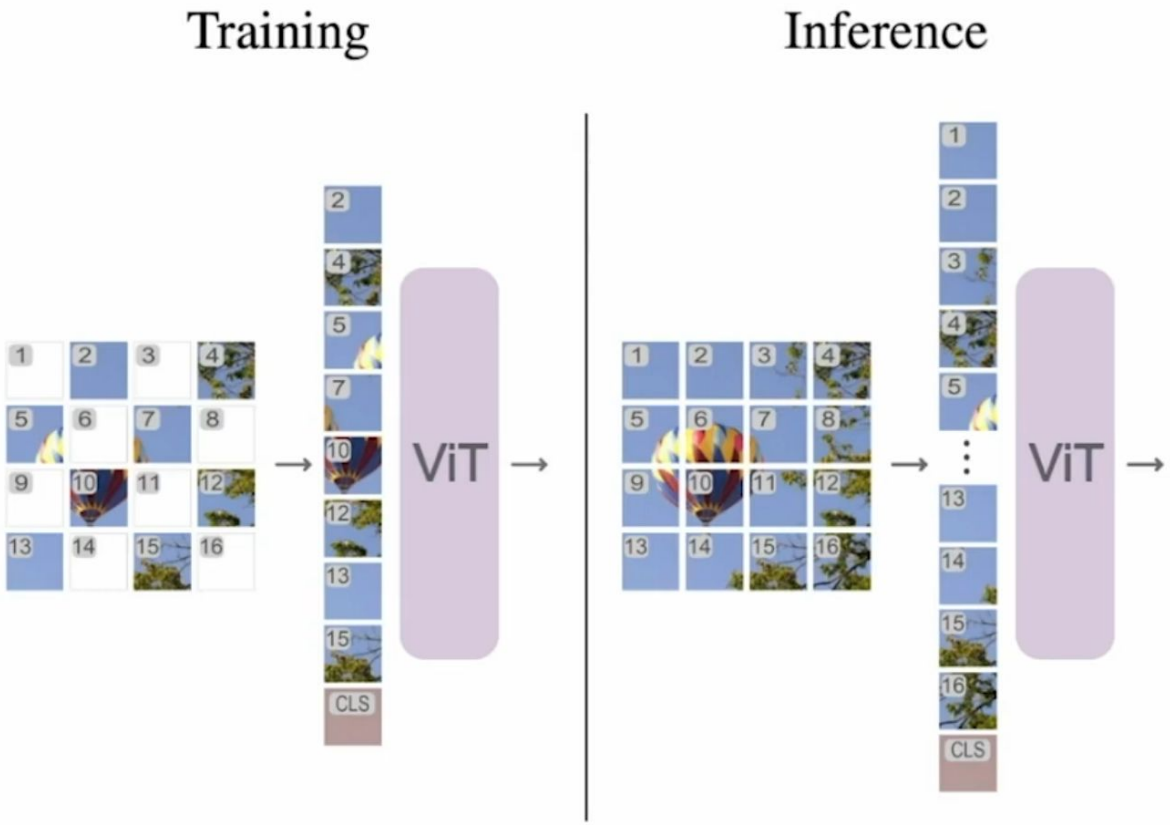
- Partial views retain core semantics

Dynamic & Random Dropping

- Dropped patches vary in every training epoch
- Eventually sees 100% of the image

Strong Regularization

- Forces the model to be perfect



During **training**, patchify the image and add positional embeddings to each patch. Uniformly sample a subset of them and use them to train the model.

At **inference** time, all patches are retained.

Methods - BiomedCLIP

Training approach: contrastive

- Loss example: InfoNCE (noise contrastive estimation) loss:

$$\mathcal{L}_I^{\text{InfoNCE}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp\left(z_i^I \cdot z_+^I/\tau\right)}{\sum_{j=1,j\neq i}^{B+1}\exp(z_i^I \cdot z_j^I/\tau)}$$

- To train a text and image encoder simultaneously, symmetrize:

$$\mathcal{L}_{I\to T} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp\left(z_i^I \cdot z_i^T/\tau\right)}{\sum_{j=1}^{B}\exp(z_i^I \cdot z_j^T/\tau)}$$

$$\mathcal{L}_{T\to I} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp\left(z_i^T \cdot z_i^I/\tau\right)}{\sum_{j=1}^{B}\exp(z_i^T \cdot z_j^I/\tau)}$$

Methods - BiomedCLIP

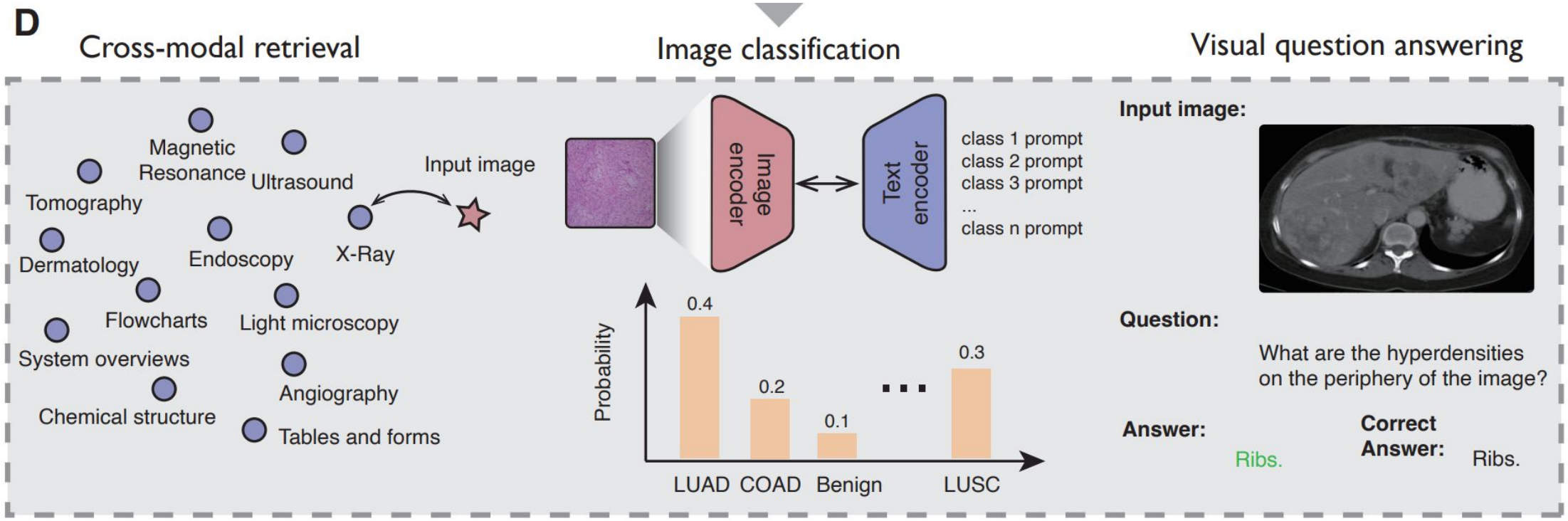
$$\mathcal{L} = -\frac{1}{2N}\left(\sum_{i=1}^{N} \log \frac{e^{\cos(\boldsymbol{I}_i, \boldsymbol{T}_i)/\tau}}{\sum_{j=1}^{N} e^{\cos(\boldsymbol{I}_i, \boldsymbol{T}_j)/\tau}} + \sum_{i=1}^{N} \log \frac{e^{\cos(\boldsymbol{I}_i, \boldsymbol{T}_i)/\tau}}{\sum_{j=1}^{N} e^{\cos(\boldsymbol{I}_j, \boldsymbol{T}_i)/\tau}}\right),$$

text encoder	vocab	context length	loss (↓)	img2txt (%) R@1(↑)	txt2img (%) R@1(↑)
GPT	50k general domain	77	0.6626	64.53	63.56
PubMedBERT	30k domain specific	77	0.5776	69.03	67.41
PubMedBERT	30k domain specific	256	**0.4807**	**73.50**	**72.26**

vision encoder	trainable params	hidden dim	loss (↓)	img2txt (%) R@1(↑)	txt2img (%) R@1(↑)
ViT-S/16	22M	384	0.5342	69.45	68.02
ViT-M/16	39M	512	0.5063	71.85	70.22
ViT-B/16	86M	768	**0.4807**	**73.50**	**72.26**

Downstream Tasks

Results

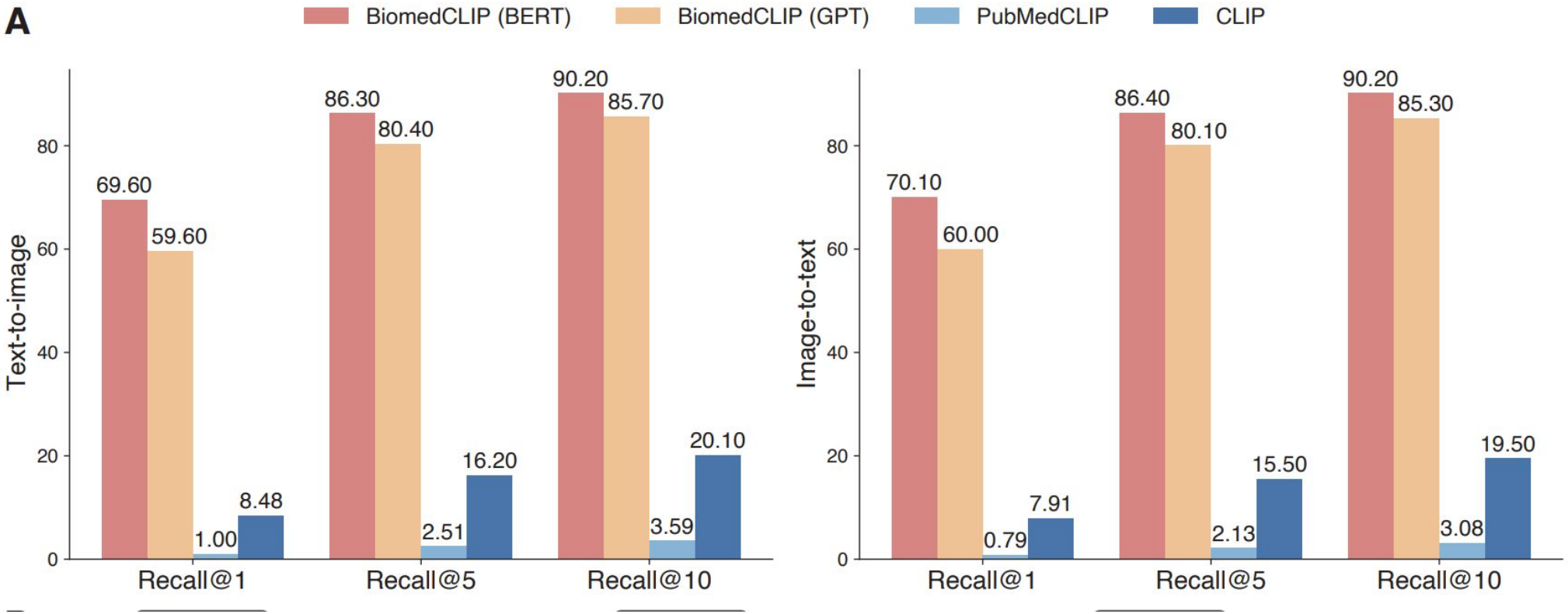

D
Cross-modal retrieval
Image classification
Visual question answering
Magnetic Resonance
Ultrasound
Tomography
Input image
Dermatology
Endoscopy
X-Ray
Flowcharts
Light microscopy
System overviews
Angiography
Chemical structure
Tables and forms
Image encoder
Text encoder
class 1 prompt
class 2 prompt
class 3 prompt
...
class n prompt
Probability
0.4
0.2
0.1
0.3
LUAD
COAD
Benign
LUSC
Input image:
Question:
What are the hyperdensities on the periphery of the image?
Answer:
Ribs.
Correct Answer:
Ribs.

Results

Task	Dataset	Metric	Description	Data size		
				Train	Dev	Test
Cross-Modal Retrieval	PMC-15M	Recall@k	Given textual description (caption), retrieve the corresponding image, or vice versa. (Image size: see **Fig. 1A**)	13.9M	13.6k	726k
Image Classification	PCam	Accuracy	Binary classification on whether a histopathology image of lymph node contains metastatic tumor tissue. (Image size: 96×96)	262,144	32,768	32,768
	LC25000 (Lung)	Accuracy	Ternary classification (benign, adenocarcinoma, squamous cell carcinoma) on histopathology images of lung tissue. (Image size: 768×768)	-	-	15,000
	LC25000 (Colon)	Accuracy	Binary classification (benign, adenocarcinoma) on histopathology images of colon tissue. (Image size: 768×768)	-	-	10,000
	TCGA-TIL	AUROC	Binary classification on whether lung H&E whole-slide image patches show adenocarcinoma. (Image size: 512×512)	-	-	2,480
	RSNA	Accuracy	Binary classification on whether chest X-rays show pneumonia. (Image size: 500×500)	18,678	4,003	9,069
VQA	VQA-RAD	Accuracy	Answer clinician questions about radiology images.	3,064	-	451
	SLAKE	Accuracy	Answer clinician questions about radiology images (X-rays, and single slices of CTs and MRIs).	4,919	1,053	1,061

Results - Cross-modal Retrieval

Retrieve the corresponding image from the caption (text-to-image retrieval) or vice versa (image-to-text retrieval)

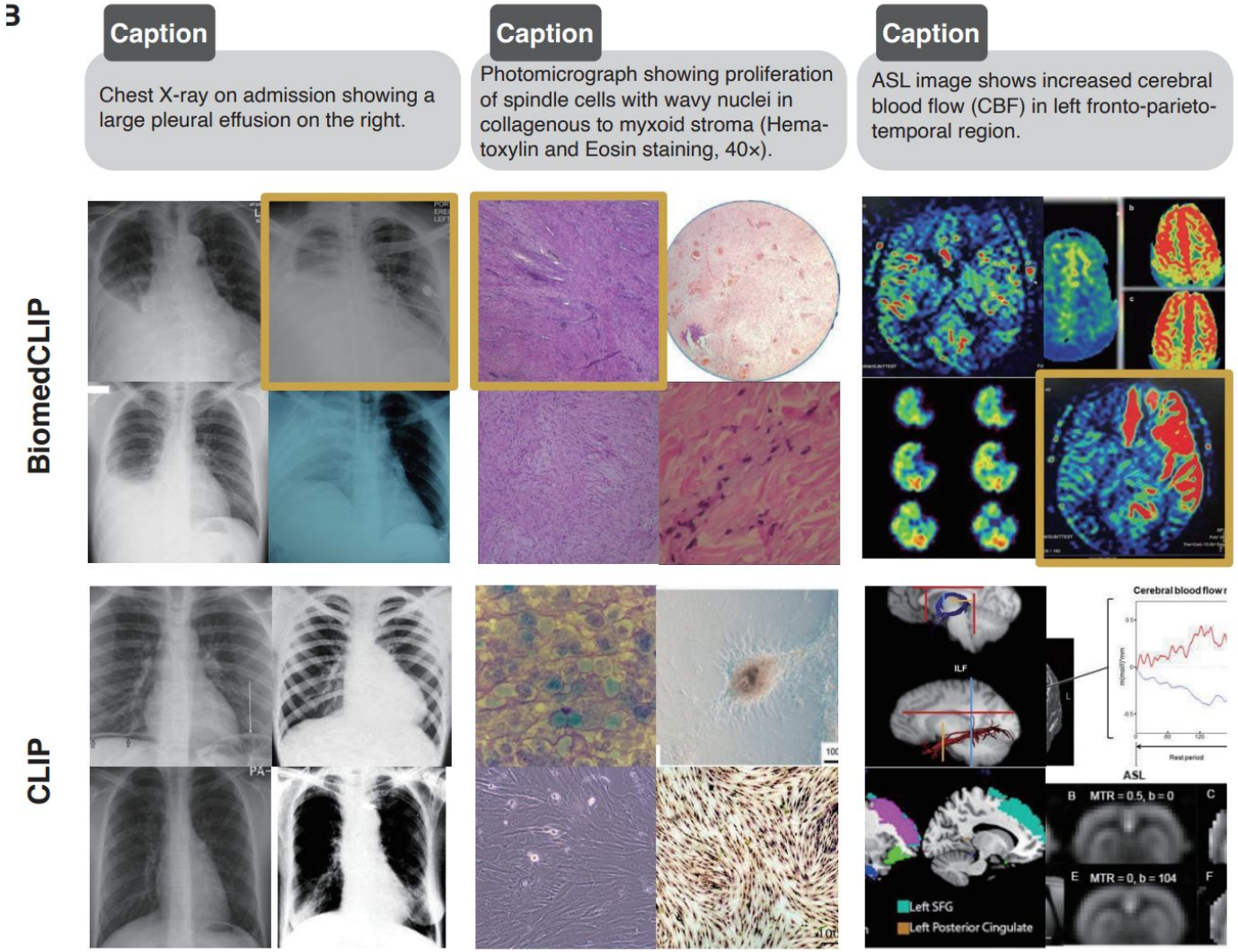


Results - Cross-modal Retrieval

3 examples comparing BiomedCLIP and CLIP on text-to-image retrieval for sample PMC captions (top-4 predictions)

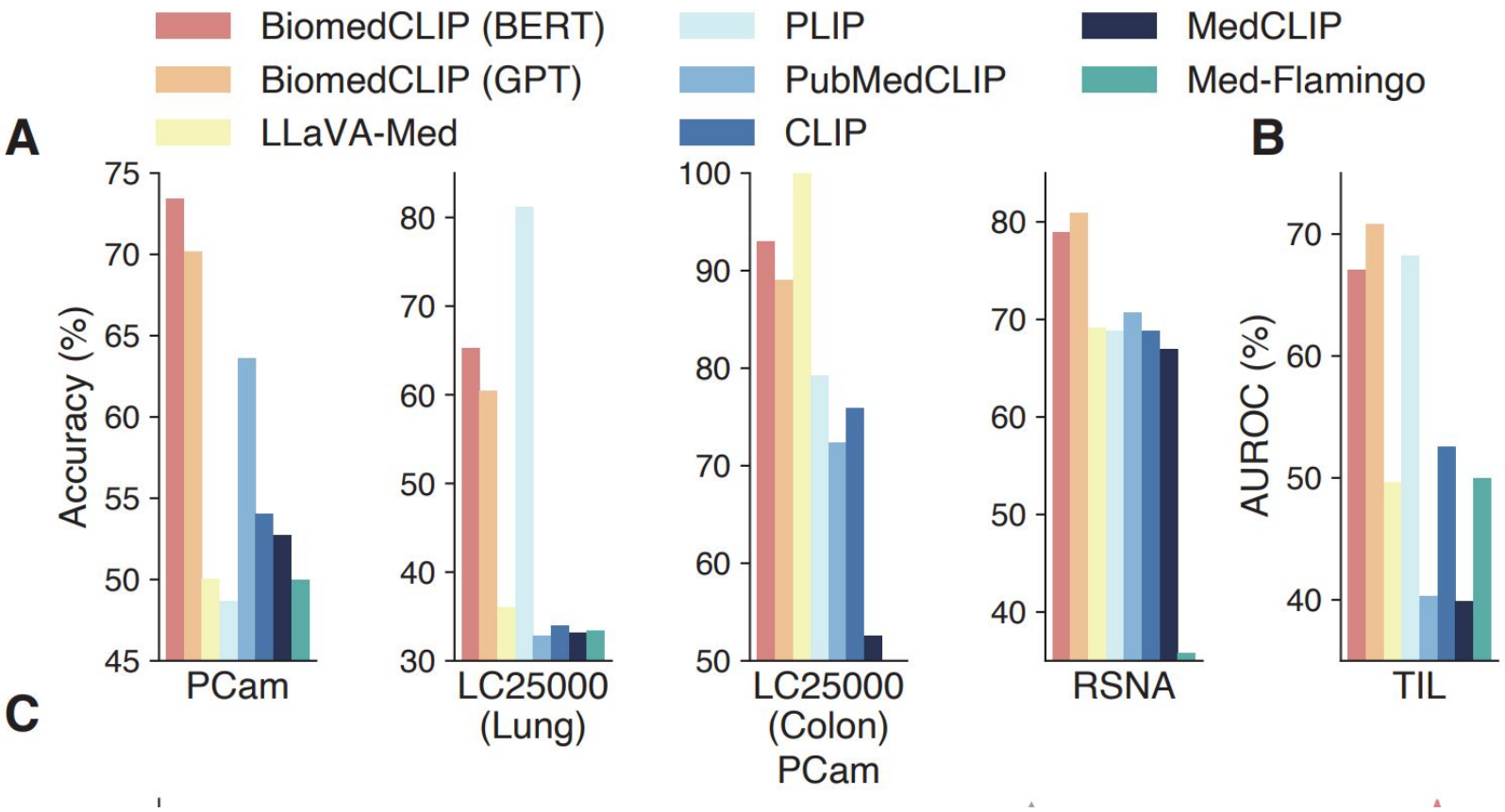
Gold box indicates the ground truth for caption

The ability attributes to large-scale pre-training



Results - Image Classification

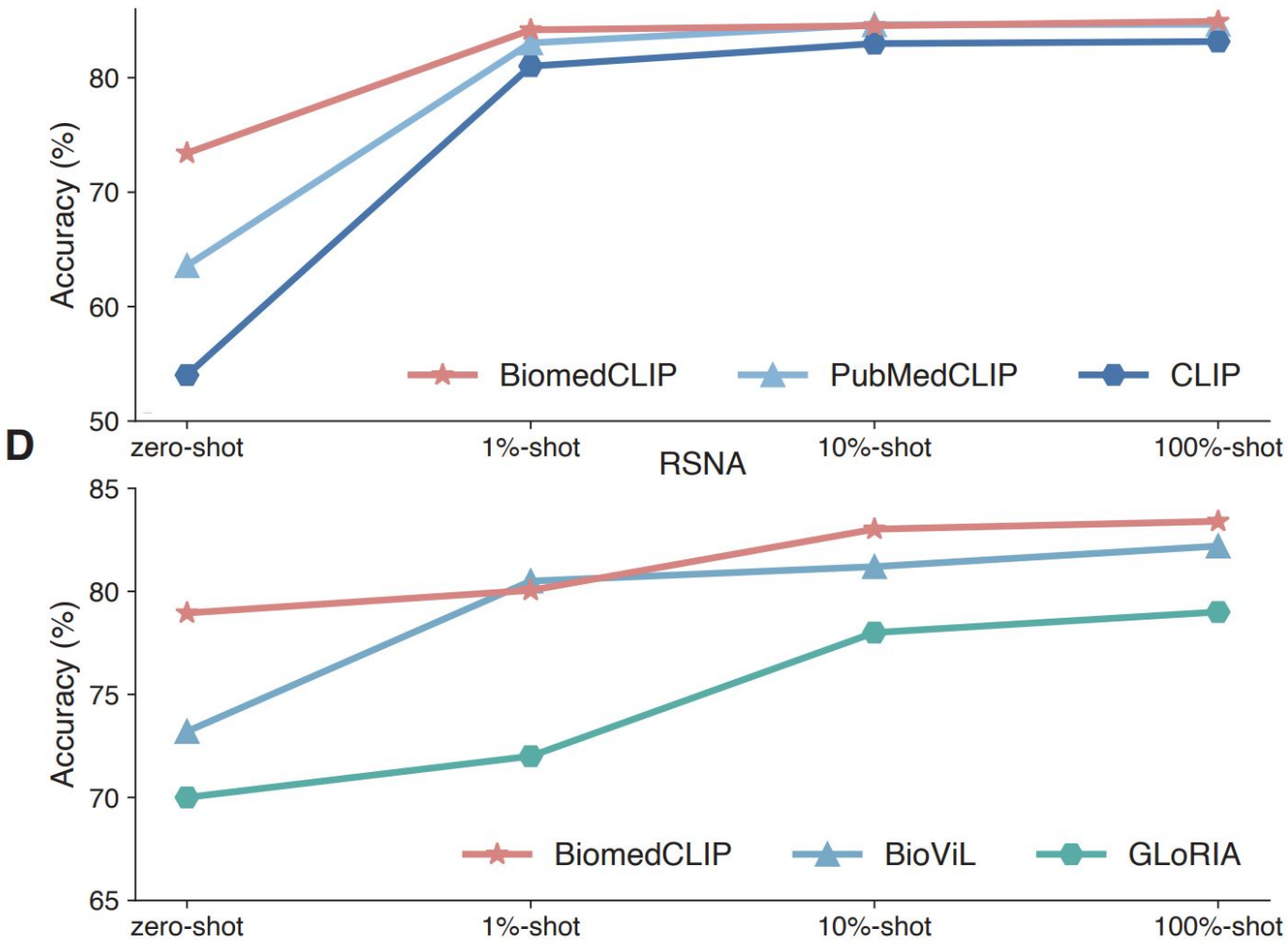
Zero-shot Performance



Results - Image Classification

Fine-tuned (linear probing) test accuracy on PCam (C) and RSNA (D)

GLoRIA and BioViL are SOTA radiology-specific models

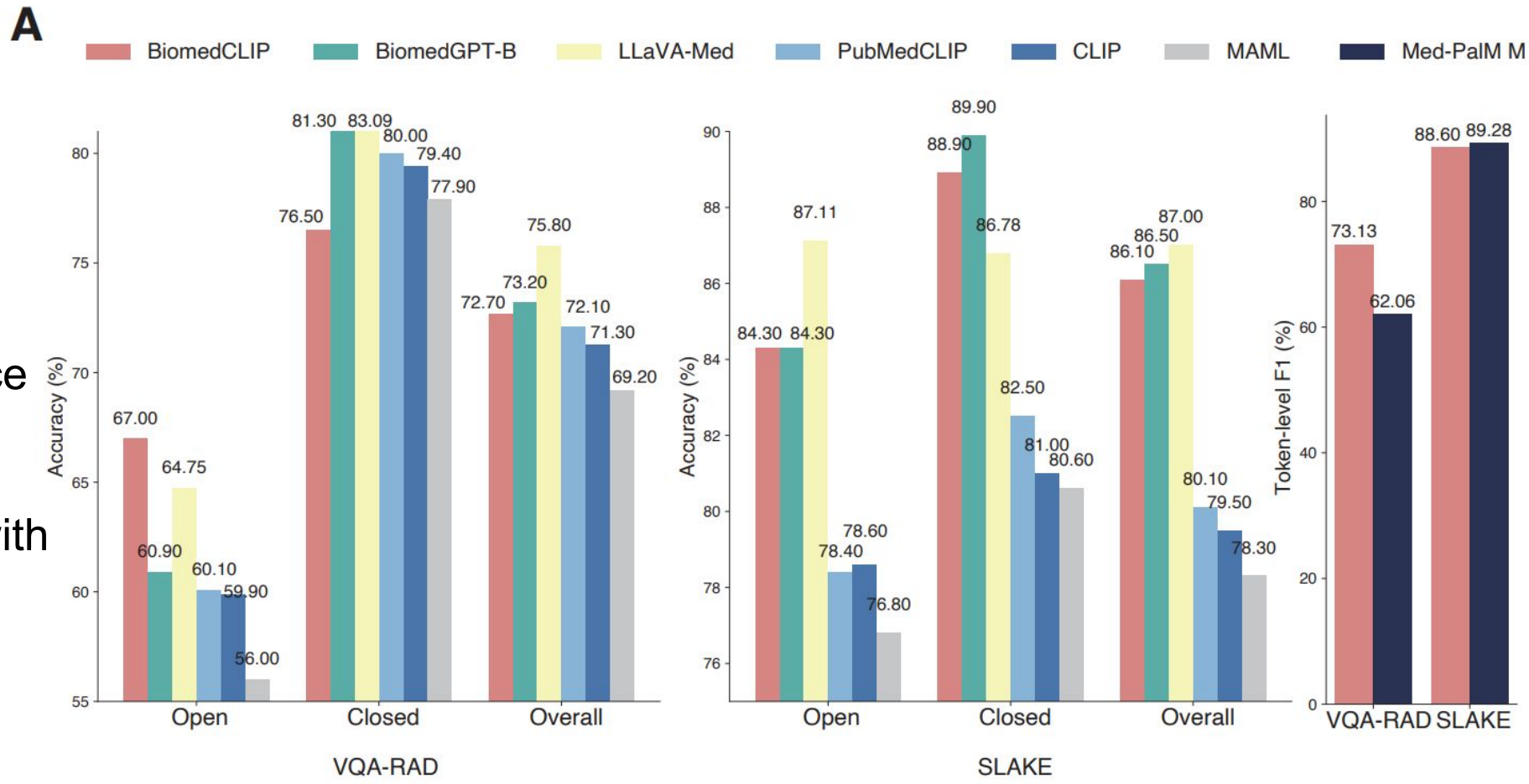


Results – Medical Visual Question Answering

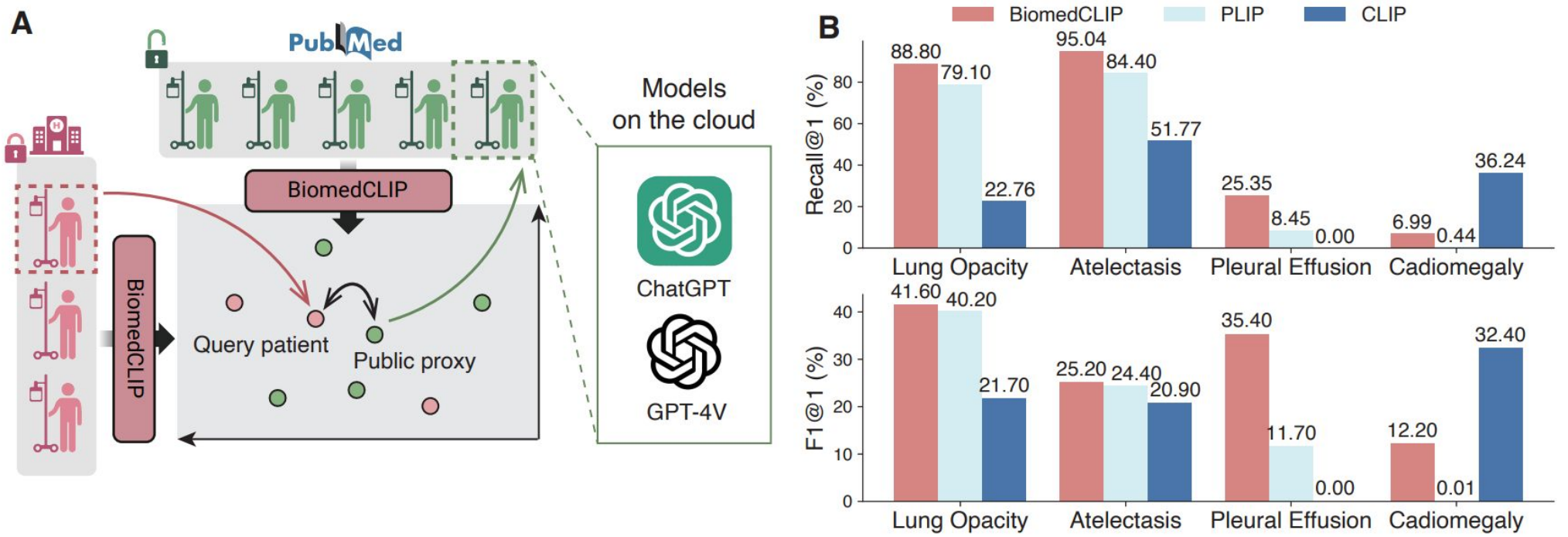
Open: open-ended answers

Closed: multiple choice (mostly yes and no)

BiomedCLIP on par with Med-PaLM M (562B) and the instruction-aware BiomedGPT-B



Results – Privacy-preserving analysis



Lower recall rate than CLIP for cardiac hypertrophy (Figure 5B)
May be limited by the title descriptions in scientific papers

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